Task 3: Sales Prediction Using Machine Learning

1. Objective

Build and compare regression models that predict Gross Sales Amount from historical retail order information, engineer calendar and past-sales features, evaluate on a held-out test period, and tune the best tree-based model.

2. Dataset and target

The source cleaned retail dataset initially contained 1,000 rows and 17 columns. Gross Sales Amount was defined as Quantity × Unit Price (INR). After excluding records with missing/invalid Order Date or target, 971 records were available for modelling. The target mean was ₹31,070.36, median ₹3,809.64, and maximum ₹4,000,000.00.

To avoid target leakage / a tautological model, Quantity and Unit Price (INR) were excluded from predictors even though they mathematically determine the target. Order ID, customer name, and coupon code were also excluded as identifiers or high-cardinality text.

3. Feature engineering and preparation

Extracted year, month, quarter, day of week, day of month, and weekend flag. Created lag-1, lag-7, 7-record rolling mean, and 30-record rolling mean using a one-row shift so current target values are not used in their own features. Rows were sorted by Order Date and Order ID. Numeric missing values were median-imputed and categorical values mode-imputed inside a Scikit-learn pipeline; categorical variables were one-hot encoded. The split used the earliest 80% for training and latest 20% for testing.

4. Model comparison

Model	RMSE (INR)	MAE (INR)	R-squared
Linear Regression	187,433.36	61,030.57	-0.0975
Tuned Random Forest	188,813.12	47,938.88	-0.1137
Random Forest	211,334.93	57,326.56	-0.3953

RMSE penalizes larger errors, MAE is the average absolute prediction error in INR, and R-squared indicates the proportion of target variance explained relative to a mean baseline. Lower RMSE/MAE and higher R-squared are generally preferred.

5. Hyperparameter tuning

RandomizedSearchCV tested 10 Random Forest configurations with 3-fold TimeSeriesSplit cross-validation using negative RMSE as the selection score. Best parameters: {'model__n_estimators': 350, 'model__min_samples_split': 5, 'model__min_samples_leaf': 4, 'model__max_features': 0.6, 'model__max_depth': 16}

6. Visualizations

Figure 1 compares actual test-set gross sales with predictions. Points closer to the diagonal indicate more accurate predictions. Large deviations indicate under- or over-prediction, especially for extreme-value transactions.

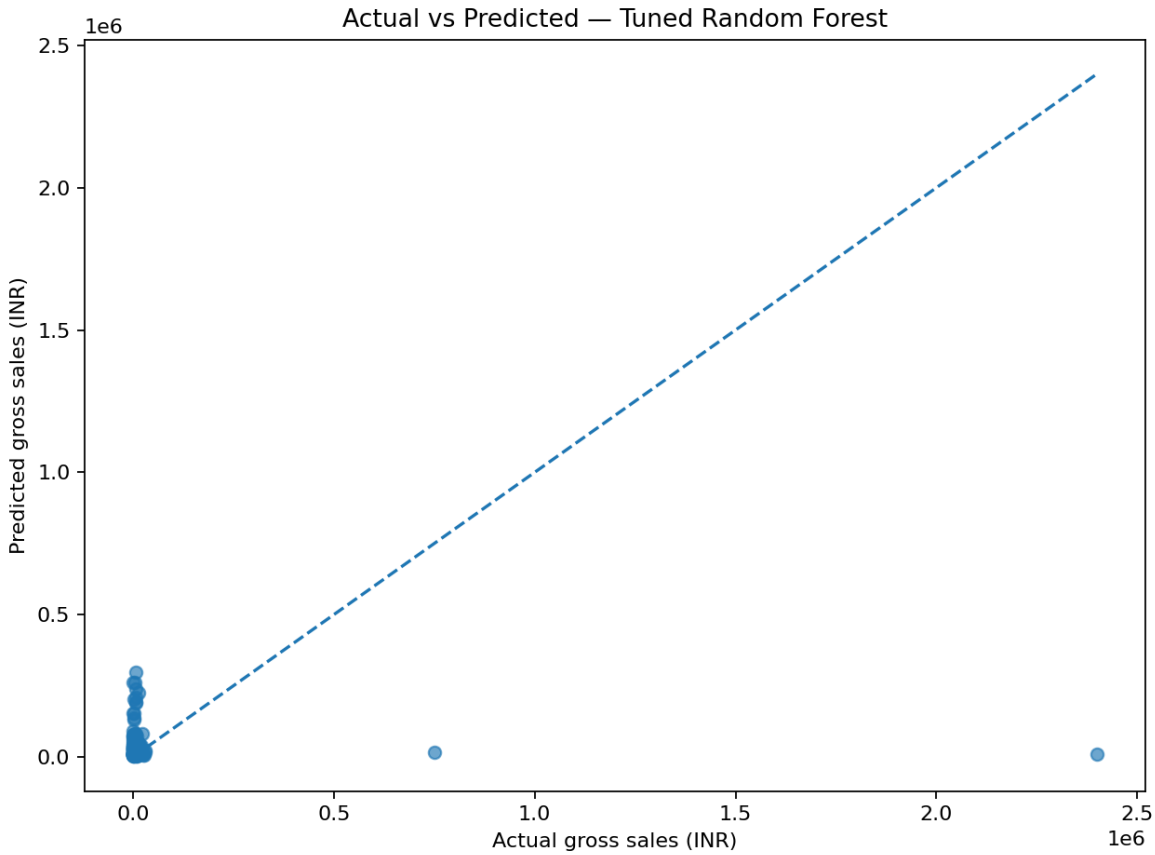


Figure 2 shows the top 15 model feature importances. These describe the features most used by the fitted Random Forest; they do not establish causation.

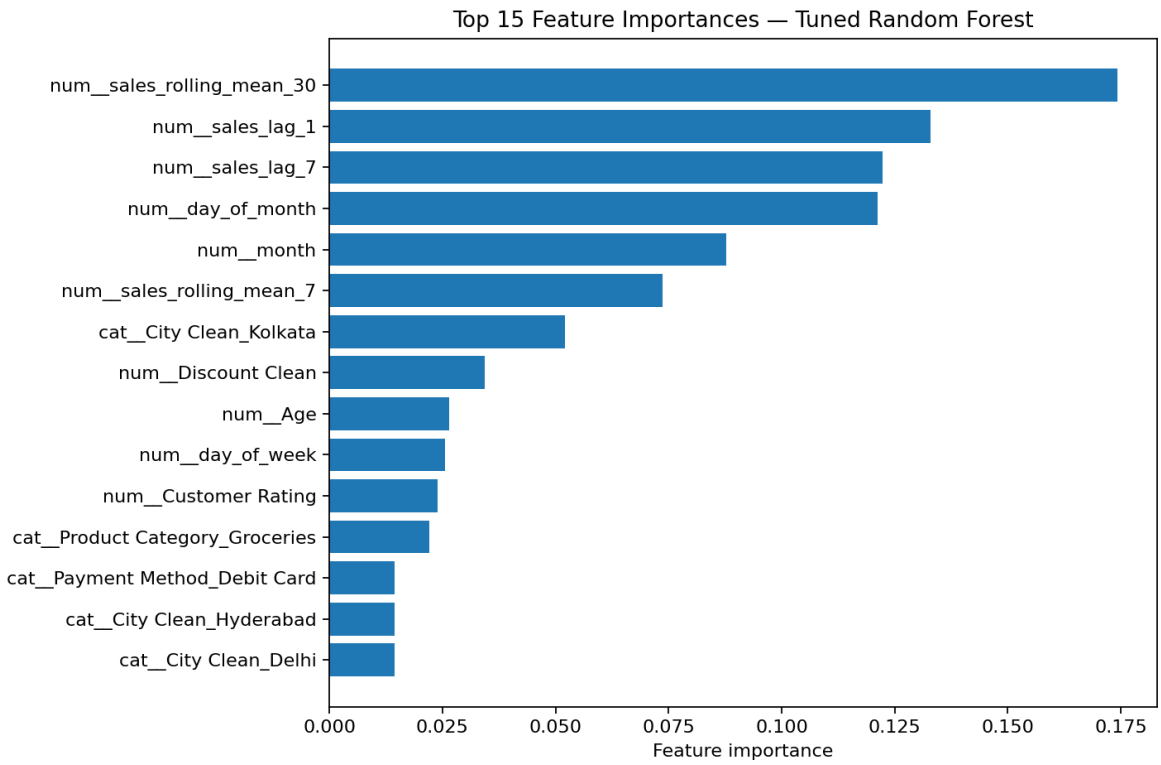
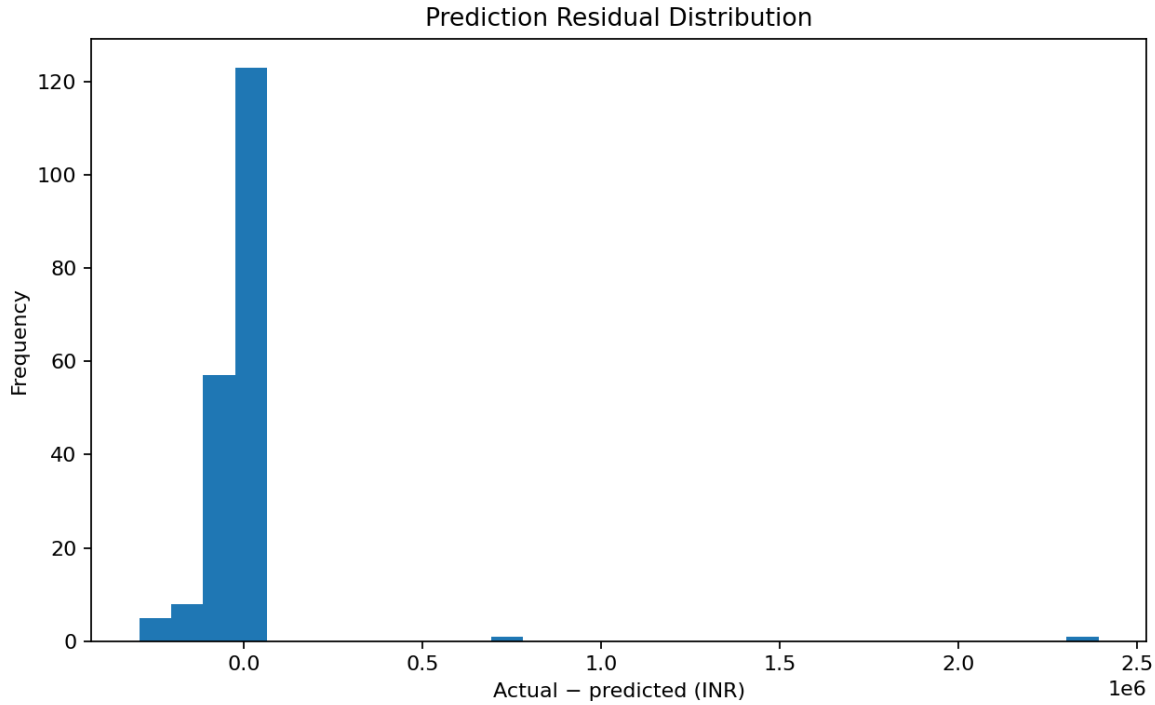


Figure 3 shows residuals (actual minus predicted). A wide distribution or long tails indicate substantial errors or difficult outlier transactions.



7. Business interpretation

The lowest test RMSE in this run was from Linear Regression (RMSE ₹187,433.36; MAE ₹61,030.57; R^2 -0.0975). Use these metrics together: RMSE is sensitive to very large errors, while MAE is easier to interpret as the typical absolute error. The model may support rough sales estimation when order attributes and prior transactions are available, but predictions for high-value outliers should be reviewed carefully.

8. Limitations and next steps

- The target is calculated from Quantity × Unit Price; if both are known, arithmetic is more reliable than a learned prediction. They were excluded to make the prediction problem non-trivial.
- The dataset covers only 2025 and is relatively small; results may not generalize to future periods or different businesses.
- Lag features are based on prior rows rather than daily aggregated sales; a production forecast should aggregate by day/week and use a time-aware backtest.
- Extreme sales values can dominate RMSE and may need business validation. Report metrics both overall and by sales-value segment in future iterations.
- RandomizedSearchCV was limited to 10 parameter combinations to keep runtime reasonable.

9. Conclusion

The project implements an end-to-end regression workflow: data preparation, date and lag feature engineering, chronological 80/20 split, Linear Regression and Random Forest comparison, RMSE/MAE/R^2 evaluation, feature importance, actual-versus-predicted visualization, residual analysis, and hyperparameter tuning. The model should be treated as an experimental baseline until validated on additional future data.